A Consolidated Overview of Datasets and Benchmarks for Machine Learning and Computer Vision

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Abstract

This manuscript provides a consolidated, practitioner-oriented overview of widely used datasets and benchmarks for machine learning (ML) and computer vision (CV). It aims to serve as a compact entry point for selecting suitable benchmarks, understanding typical evaluation settings, and recognizing common dataset characteristics such as modality, scale, and annotation scope. Beyond listing canonical datasets, the manuscript highlights benchmark hubs and repositories that standardize access, splits, and metadata, and it emphasizes how dataset choice influences comparability, reproducibility, and the validity of conclusions. The overview is organized along two broad domains—ML and CV—and covers representative task families including tabular learning, time series, graphs, recommender systems, natural language processing, reinforcement learning, and multiple CV subareas (classification, detection, segmentation, 3D vision, remote sensing, vision–language, and medical imaging). The manuscript is intended as a living reference that supports both newcomers and experienced researchers in rapidly mapping the benchmark landscape and in making transparent, well-motivated dataset selections for experiments and reporting.

Table of Contents

1 Introduction and Motivation **6**

2 Machine Learning **7**

2.1 Benchmark hubs and repositories . . . 8
2.1 OpenML . . . 8
2.2 Hugging Face Datasets . . . 8
2.3 TensorFlow Datasets (TFDS) . . . 8
2.4 Kaggle Datasets . . . 8
2.2 Tabular ML (classical supervised learning) . . . 8
2.5 UCI ML Repository . . . 8
2.6 Adult (Census Income) . . . 8
2.7 Bank Marketing . . . 8
2.8 HIGGS . . . 8
2.9 Credit Card Fraud (EU) . . . 8
2.3 Time series and forecasting . . . 8
2.10 UCR Time Series Classification Archive . . . 8
2.11 UEA Multivariate TSC Archive (2018) . . . 8
2.12 M4 . . . 8
2.13 M5 . . . 8
2.14 ETTh/ETTm . . . 8
2.4 Anomaly detection (general) . . . 8
2.15 UNSW-NB15 . . . 9
2.16 Numenta Anomaly Benchmark (NAB) . . . 9
2.5 Graph machine learning . . . 9
2.17 Open Graph Benchmark (OGB) . . . 9
2.18 Cora . . . 9
2.19 CiteSeer . . . 9
2.20 Reddit (Graph) . . . 9
2.6 Recommender systems . . . 9
2.21 MovieLens . . . 9
2.22 Amazon Reviews . . . 9
2.7 NLP benchmarks (understanding, QA, IE) . . . 9
2.23 GLUE . . . 9
2.24 SuperGLUE . . . 9
2.25 SQuAD . . . 9
2.26 MS MARCO . . . 9
2.27 OntoNotes 5.0 . . . 9
2.28 XNLI . . . 9
2.29 TyDi QA . . . 9
2.8 NLP pretraining corpora . . . 9
2.30 C4 . . . 9
2.31 The Pile . . . 10
2.32 BookCorpus . . . 10
2.9 Machine translation and multilingual corpora . . . 10
2.33 OPUS . . . 10
2.34 FLORES-200 . . . 10

2.10 Speech and audio . . . 10
2.35 LibriSpeech . . . 10
2.36 Common Voice . . . 10
2.37 AudioSet . . . 10
2.38 VoxCeleb . . . 10
2.11 Reinforcement learning (environments and offline RL) . . . 10
2.39 Arcade Learning Environment (ALE) . . . 10
2.40 D4RL . . . 10
2.41 DeepMind Control Suite . . . 10
2.42 MuJoCo Benchmarks . . . 10
2.12 Healthcare / EHR and clinical prediction . . . 10
2.43 MIMIC-III . . . 10
2.44 MIMIC-IV . . . 10
2.45 eICU Collaborative Research Database . . . 10
2.13 Medical imaging (ML-relevant benchmarks) . . . 11
2.46 NIH ChestX-ray14 . . . 11
2.47 CheXpert . . . 11
2.48 MIMIC-CXR . . . 11
2.49 BraTS . . . 11
2.14 Fairness, robustness, and shift evaluation . . . 11
2.50 Waterbirds . . . 11

3 Computer Vision **12**
3.1 Image classification (general) . . . 13
3.1 MNIST . . . 13
3.2 Fashion-MNIST . . . 13
3.3 CIFAR-10/100 . . . 13
3.4 SVHN . . . 13
3.5 ImageNet . . . 13
3.6 ImageNet-21k . . . 13
3.2 Fine-grained recognition . . . 13
3.7 CUB-200-2011 . . . 13
3.8 Stanford Cars . . . 13
3.9 FGVC Aircraft . . . 13
3.10 iNaturalist . . . 13
3.3 Object detection . . . 13
3.11 PASCAL VOC . . . 13
3.12 MS COCO . . . 13
3.13 Open Images . . . 13
3.14 Objects365 . . . 13
3.4 Instance segmentation . . . 13
3.15 COCO (Instance Segmentation) . . . 13
3.16 LVIS . . . 13
3.17 ADE20K (parts/instances) . . . 13
3.5 Semantic segmentation . . . 14
3.18 ADE20K . . . 14
3.19 Cityscapes . . . 14

3.20 PASCAL-Context . . . 14
3.21 COCO-Stuff . . . 14
3.6 Human pose estimation and keypoints . . . 14
3.22 MPII Human Pose . . . 14
3.23 COCO Keypoints . . . 14
3.24 Human3.6M . . . 14
3.25 PoseTrack . . . 14
3.7 Face analysis . . . 14
3.26 WIDER FACE . . . 14
3.27 CelebA . . . 14
3.28 LFW . . . 14
3.29 FFHQ . . . 14
3.8 Tracking and MOT . . . 14
3.30 MOT17 . . . 14
3.31 MOT20 . . . 14
3.32 LaSOT . . . 14
3.9 Video action recognition . . . 14
3.33 Kinetics . . . 14
3.34 HMDB51 . . . 14
3.35 UCF101 . . . 14
3.36 Something-Something V2 . . . 15
3.10 Video segmentation . . . 15
3.37 DAVIS . . . 15
3.38 YouTube-VOS . . . 15
3.11 Optical flow and correspondence . . . 15
3.39 Middlebury Flow . . . 15
3.40 MPI-Sintel . . . 15
3.41 FlyingChairs . . . 15
3.42 FlyingThings3D . . . 15
3.12 Stereo and depth estimation . . . 15
3.43 KITTI . . . 15
3.44 KITTI Depth . . . 15
3.45 NYU Depth v2 . . . 15
3.46 SUN RGB-D . . . 15
3.13 Autonomous driving (large-scale) . . . 15
3.47 BDD100K . . . 15
3.48 Waymo Open Dataset . . . 15
3.49 nuScenes . . . 15
3.14 3D vision (scans, reconstruction, point clouds) . . . 15
3.50 ScanNet . . . 15
3.51 ShapeNet . . . 15
3.52 ModelNet40 . . . 15
3.53 S3DIS . . . 16
3.15 Remote sensing and aerial imagery . . . 16
3.54 DOTA . . . 16
3.55 xView . . . 16
3.56 DeepGlobe . . . 16

3.16 Document AI / OCR . . . 16
3.57 IAM Handwriting . . . 16
3.58 PubTabNet . . . 16
3.17 Vision–language (image–text) . . . 16
3.59 Visual Genome . . . 16
3.60 VQA . . . 16
3.61 Conceptual Captions (CC3M) . . . 16
3.62 Conceptual Captions (CC12M) . . . 16
3.63 Flickr30k . . . 16
3.64 MSR-VTT . . . 16
3.65 LAION-5B . . . 16
3.18 Generative modeling and super-resolution . . . 16
3.66 CelebA-HQ . . . 16
3.67 DIV2K . . . 16
3.68 FFHQ . . . 16
3.19 Robustness and distribution shift . . . 16
3.69 ImageNet-A . . . 17
3.70 ImageNet-R . . . 17
3.71 ImageNet-Sketch . . . 17
3.72 Stylized-ImageNet . . . 17
3.20 Medical imaging (CV-specific) . . . 17
3.73 BraTS . . . 17
3.74 LUNA16 . . . 17
3.75 DRIVE . . . 17

4 Conclusion and Outlook **18**

1 Introduction and Motivation

Datasets and benchmarks are central to progress in machine learning and computer vision: they operationalize tasks, define evaluation protocols, and enable quantitative comparisons across methods. At the same time, dataset selection is a frequent source of hidden variance in experimental outcomes. Differences in label definitions, sampling strategies, train/validation/test splits, annotation noise, and domain coverage can easily outweigh architectural or optimization improvements. Consequently, responsible empirical work requires more than picking a popular dataset. It requires understanding what a benchmark measures, what it omits, and how its assumptions align with the intended deployment setting.

The benchmark landscape has expanded rapidly, spanning classical tabular problems, modern web-scale corpora for foundation models, specialized medical and scientific datasets, and increasingly multimodal benchmarks. In parallel, dataset access has shifted toward standardized repositories and loaders that facilitate reproducibility through versioning, checksums, and canonical splits. This manuscript consolidates representative datasets and benchmark suites across ML and CV into structured tables that summarize modality, approximate scale, and typical tasks. The goal is not exhaustiveness, but coverage of widely referenced anchors and practical starting points for experiment design, baselining, and reporting.

Throughout the manuscript, the term dataset refers to a curated collection of samples (e.g., images, time series, text, or graphs) with accompanying metadata and often annotations, while benchmark refers to a dataset paired with an evaluation protocol (metrics, splits, constraints, and reporting conventions). The presented overview is meant to be updated over time and to support transparent documentation practices, e.g., explicitly stating dataset versions, preprocessing steps, and evaluation protocols.

2 Machine Learning

Machine learning benchmarks span a broad range of modalities and task formulations. Classical supervised learning on tabular data remains important for structured prediction in domains such as finance, marketing, and healthcare, while time series benchmarks capture forecasting and anomaly detection challenges under temporal dependencies and seasonality. Graph benchmarks support research on relational reasoning and representation learning for networks, and recommender-system datasets highlight issues such as sparsity, popularity bias, and temporal drift. In natural language processing (NLP), benchmark suites for understanding and question answering (QA) coexist with massive pre-training corpora that fuel modern large language models (LLM), making it important to distinguish evaluation benchmarks from pretraining data sources. Reinforcement learning (RL) benchmarks often come as environment suites or logged trajectories, and their evaluation depends strongly on interaction protocols (online vs. offline RL).

A recurring theme across ML benchmarks is the interplay between data scale, annotation quality, and evaluation stability. Small and well-curated datasets can support controlled comparisons but may saturate quickly or fail to reflect real-world variability. Large-scale datasets improve statistical power but increase the risk of dataset-specific shortcuts, spurious correlations, and untracked distribution shift. Moreover, some benchmarks are primarily historical (useful for continuity) while others are designed to stress robustness, fairness, or out-of-distribution generalization. For transparent reporting, it is good practice to document dataset versions, canonical splits, leakage checks, and any filtering or deduplication steps, especially when benchmarks are accessed via evolving repositories.

Subsequently, the following abbreviations are used:

- ASR = automatic speech recognition
- EHR = electronic health records
- GNN = graph neural network
- HEP = high-energy physics
- ICU = intensive care unit
- IR = information retrieval
- MedImg = medical imaging
- MRI = magnetic resonance imaging
- MT = machine translation
- Multivar = multivariate
- NLP = natural language processing
- QA = question answering
- RL = reinforcement learning
- TSC = time series classification

Table 1: Machine learning datasets.

Dataset	Area	Modality	Scale	Typical tasks / notes
2.1 Benchmark hubs and repositories				
OpenML [1]	Tabular	Mixed	10k+ tasks	Benchmarking hub with standardized tasks, splits, and metadata
Hugging Face Datasets [2]	NLP/Multi	Mixed	1000s	Unified dataset hosting/loader; many tasks and modalities
TensorFlow Datasets (TFDS) [3]	Multi	Mixed	100s	Curated dataset loaders with versioning and checksums
Kaggle Datasets [4]	Multi	Mixed	100k+	Community datasets; quality varies; good for rapid prototyping
2.2 Tabular ML (classical supervised learning)				
UCI ML Repository [5]	Tabular	Tabular	100s	Classic ML datasets; heterogeneous difficulty and documentation
Adult (Census Income) [6]	Tabular	Tabular	48k	Binary classification; fairness discussions common
Bank Marketing [7]	Tabular	Tabular	45k	Marketing response prediction; imbalanced classification
HIGGS [8]	Tabular	Tabular	11M	Large-scale binary classification; HEP-inspired
Credit Card Fraud (EU) [9]	Tabular	Tabular	285k	Highly imbalanced anomaly/fraud detection
2.3 Time series and forecasting				
UCR Time Series Classification Archive [10]	Time series	1D	100+ sets	Univariate time series classification benchmarks
UEA Multivariate TSC Archive (2018) [11]	Time series	Multivar	30+ sets	Multivariate time series classification benchmarks
M4 [12]	Forecast	Time series	100k series	Large forecasting competition dataset
M5 [13]	Forecast	Time series	Walmart	Retail demand forecasting; hierarchical aggregation
ETTh/ETTm [14]	Forecast	Time series	2 years	Transformer time series benchmarks (ETT family)

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Dataset	Area	Modality	Scale	Typical tasks / notes
2.4 Anomaly detection (general)				
UNSW-NB15 [15]	Anomaly	Tabular	2.5M	Modern intrusion detection benchmark alternative
Numenta Anomaly Benchmark (NAB) [16]	Anomaly	Time series	58 files	Streaming anomaly detection with labeled windows
2.5 Graph machine learning				
Open Graph Benchmark (OGB) [17]	Graphs	Graph	Many	Standardized graph benchmarks and evaluation protocols
Cora [18]	Graphs	Citation graph	2.7k nodes	Node classification; classic GNN benchmark
CiteSeer [19]	Graphs	Citation graph	3.3k nodes	Node classification; classic benchmark
Reddit (Graph) [20]	Graphs	Graph	232k nodes	Inductive node classification; community graphs
2.6 Recommender systems				
MovieLens [21]	Recsys	Ratings	Various	Collaborative filtering; multiple sizes/versions
Amazon Reviews [22]	Recsys/NLP	Text+ratings	Large	Review-based recommendation; domain shift across categories
2.7 NLP benchmarks (understanding, QA, IE)				
GLUE [23]	NLP	Text	9 tasks	General language understanding evaluation suite
SuperGLUE [24]	NLP	Text	8 tasks	Harder successor to GLUE
SQuAD [25]	NLP	Text	100k+ QAs	Reading comprehension / extractive QA
MS MARCO [26]	IR/QA	Text	Large	Passage ranking, retrieval, QA-style supervision
OntoNotes 5.0 [27]	NLP	Text	Large	NER, coreference; richer annotation
XNLI [28]	NLP	Text	15 langs	Cross-lingual NLI benchmark
TyDi QA [29]	NLP	Text	11 langs	Multilingual QA across typologically diverse languages
2.8 NLP pretraining corpora				

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Dataset	Area	Modality	Scale	Typical tasks / notes
C4 [30]	NLP	Text	Large	Cleaned Common Crawl; popular for transformer pretraining
The Pile [31]	NLP	Text	800GB	Curated mixture for LLM pretraining
BookCorpus [32]	NLP	Text	Large	Books; used in early transformer pretraining
2.9 Machine translation and multilingual corpora				
OPUS [33]	MT	Text	Large	Open parallel corpora collection; many domains/languages
FLORES-200 [34]	MT/NLP	Text	200 langs	Many-to-many MT evaluation benchmark
2.10 Speech and audio				
LibriSpeech [35]	Speech	Audio	1000h	ASR benchmark (audiobooks)
Common Voice [36]	Speech	Audio	Large	Multilingual ASR; community-recorded, noisy conditions
AudioSet [37]	Audio	Audio	2M clips	Audio event classification with ontology-based labels
VoxCeleb [38]	Speech	Audio/Video	Large	Speaker verification/identification in the wild
2.11 Reinforcement learning (environments and offline RL)				
Arcade Learning Environment (ALE) [39]	RL	Games	57 games	Atari-style RL benchmark suite
D4RL [40]	RL	Trajectories	Many	Offline RL datasets + standardized evaluation
DeepMind Control Suite [41]	RL	Sim	Many tasks	Continuous control benchmarks
MuJoCo Benchmarks [42]	RL	Sim	Many tasks	Locomotion/manipulation; common in policy learning
2.12 Healthcare / EHR and clinical prediction				
MIMIC-III [43]	EHR	Structured + notes	Large	ICU EHR benchmark; many clinical prediction tasks
MIMIC-IV [44]	EHR	Structured + notes	Large	Newer ICU EHR release

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Dataset	Area	Modality	Scale	Typical tasks / notes
eICU Collaborative Research Database [45]	EHR	Structured	Large	Multi-center ICU EHR; diagnosis/outcome prediction
2.13 Medical imaging (ML-relevant benchmarks)				
NIH ChestX-ray14 [46]	MedImg	X-ray	112k	Multi-label thoracic disease classification
CheXpert [47]	MedImg	X-ray	224k	Chest radiograph labels; uncertain labels handling
MIMIC-CXR [48]	MedImg	X-ray	370k	Large chest X-ray + reports; multimodal learning
BraTS [49]	MedImg	MRI	2k+	Brain tumor segmentation benchmark
2.14 Fairness, robustness, and shift evaluation				
Waterbirds [50]	Robust	Images	~10k	Spurious correlation benchmark (background vs. label)

3 Computer Vision

Computer vision benchmarks are typically organized by task families such as classification, object detection, segmentation, tracking, and 3D perception. While early datasets were relatively small and curated, modern CV research increasingly relies on large-scale, diverse datasets and on standardized evaluation protocols (e.g., mean average precision for detection, intersection-over-union for segmentation, or benchmark-specific leaderboards). The community also distinguishes between datasets intended for supervised evaluation and datasets primarily used for representation learning or pretraining. In addition, multimodal vision–language datasets have become prominent as foundation models blur the boundary between CV and NLP, introducing new concerns around web-scale collection, noise, and licensing constraints.

Choosing a CV benchmark requires attention to label taxonomies, annotation granularity (image-level labels vs. instance masks vs. panoptic labels), domain coverage (e.g., consumer photos, autonomous driving, or medical imaging), and the expected generalization regime. Many datasets embed biases—for instance, long-tail class frequency, geographic or demographic skew, or dataset-specific artifacts—that can lead to brittle improvements. Robustness-oriented benchmarks explicitly probe such failure modes via distribution shift, adversarial filtering, or domain transfer (e.g., renditions, sketches, or stylized images). For practical experimentation, it is also crucial to report preprocessing (e.g., image resolution and augmentation), evaluation details (e.g., IoU thresholds or per-class averaging), and any additional supervision (e.g., external pretraining), as these strongly affect comparability.

Subsequently, the following abbreviations are used:

- Ann = annotations
- Cls = classification
- CT = computed tomography
- CV = computer vision
- Det = (object) detection
- FaceAttr = face attribute recognition
- FaceDet = face detection
- FaceID = face identification/verification
- Flow = optical flow
- Gen = generative modeling
- InstSeg = instance segmentation
- LiDAR = light detection and ranging
- MedCV = medical computer vision
- MOT = multi-object tracking
- MRI = magnetic resonance imaging
- Panoptic = panoptic segmentation
- Pose = (2D) pose estimation
- Pose3D = 3D pose estimation
- RGB-D = color + depth
- SemSeg = semantic segmentation
- SR = super-resolution
- V+L = vision–language
- VidSeg = video segmentation

Table 2: Computer vision datasets.

Dataset	Area	Modality	Scale	Typical tasks / notes
3.1 Image classification (general)				
MNIST [51]	Cls	Images	70k	Handwritten digits; classic baseline
Fashion-MNIST [52]	Cls	Images	70k	Apparel classification; MNIST-style
CIFAR-10/100 [53]	Cls	Images	60k	Small natural images benchmark
SVHN [54]	Cls	Images	600k	Street numbers; digit recognition
ImageNet [55]	Cls	Images	14M+	Large-scale classification; pretraining cornerstone
ImageNet-21k [56]	Cls	Images	14M+	Larger label space; used for pretraining
3.2 Fine-grained recognition				
CUB-200-2011 [57]	Fine-grained	Images	11k	Bird species classification with part annotations
Stanford Cars [58]	Fine-grained	Images	16k	Car model classification
FGVC Aircraft [59]	Fine-grained	Images	10k	Aircraft variant classification
iNaturalist [60]	Fine-grained	Images	Large	Long-tail species recognition
3.3 Object detection				
PASCAL VOC [61]	Det	Images	20 classes	Classic detection benchmark
MS COCO [62]	Det	Images	330k	Detection + segmentation + keypoints
Open Images [63]	Det	Images	9M+	Large-scale detection with many classes
Objects365 [64]	Det	Images	2M	Large-scale object detection for diverse categories
3.4 Instance segmentation				
COCO (Instance Segmentation) [62]	InstSeg	Images	330k	Instance masks; common baseline
LVIS [65]	InstSeg	Images	1000+ cls	Long-tail instance segmentation

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Dataset	Area	Modality	Scale	Typical tasks / notes
ADE20K (parts/instances) [66]	InstSeg	Images	25k+	Scene parsing with richer label structure
3.5 Semantic segmentation				
ADE20K [66]	SemSeg	Images	25k+	Scene parsing / dense semantic labels
Cityscapes [67]	SemSeg	Images	5k fine	Urban driving scenes; strong benchmark
PASCAL-Context [68]	SemSeg	Images	10k	Dense labels derived from PASCAL
COCO-Stuff [69]	SemSeg	Images	164k	Stuff + thing labels; semantic segmentation
3.6 Human pose estimation and keypoints				
MPII Human Pose [70]	Pose	Images	25k+	2D pose estimation benchmark
COCO Keypoints [62]	Pose	Images	330k	Keypoint detection; standard evaluation
Human3.6M [71]	Pose3D	Video+3D	Large	3D pose / mocap benchmark
PoseTrack [72]	Pose	Video	Large	Pose tracking in videos
3.7 Face analysis				
WIDER FACE [73]	FaceDet	Images	32k	Face detection in the wild
CelebA [74]	FaceAttr	Images	200k	Face attributes and identities
LFW [75]	FaceID	Images	13k	Face verification; legacy benchmark
FFHQ [76]	Faces	Images	70k	High-quality face images; generation and restoration
3.8 Tracking and MOT				
MOT17 [77]	Tracking	Video	~7 seq	Multi-object tracking benchmark
MOT20 [78]	Tracking	Video	~4 seq	Crowded multi-object tracking
LaSOT [79]	Tracking	Video	1.4k	Single object long-term tracking
3.9 Video action recognition				
Kinetics [80]	Action	Video	Hundreds k	Human action recognition benchmark
HMDB51 [81]	Action	Video	7k	Action recognition; smaller benchmark

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Dataset	Area	Modality	Scale	Typical tasks / notes
UCF101 [82]	Action	Video	13k	Action recognition; classic dataset
Something-Something V2 [83]	Action	Video	220k	Fine-grained human-object interaction actions
3.10 Video segmentation				
DAVIS [84]	VidSeg	Video	Small	Video object segmentation benchmark
YouTube-VOS [85]	VidSeg	Video	Large	Large-scale video object segmentation
3.11 Optical flow and correspondence				
Middlebury Flow [86]	Flow	Pairs	Small	Classic optical flow evaluation
MPI-Sintel [87]	Flow	Video	Large	Synthetic optical flow benchmark with ground truth
FlyingChairs [88]	Flow	Pairs	22k	Synthetic flow training dataset
FlyingThings3D [89]	Flow	3D	Large	Synthetic flow + disparity; training for correspondence
3.12 Stereo and depth estimation				
KITTI [90]	Depth	RGB+LiDAR	Large	Stereo, flow, odometry, 3D detection
KITTI Depth [91]	Depth	RGB+LiDAR	Large	Depth completion/estimation tasks
NYU Depth v2 [92]	RGB-D	RGB-D	Small	Indoor RGB-D; depth + semantics
SUN RGB-D [93]	RGB-D	RGB-D	10k	Indoor scene understanding with depth
3.13 Autonomous driving (large-scale)				
BDD100K [94]	Drive	Video	100k	Driving video: detection, lane, segmentation
Waymo Open Dataset [95]	Drive	Multi-sensor	Large	3D detection/tracking from LiDAR + cameras
nuScenes [96]	Drive	Multi-sensor	1k scenes	360-degree multi-sensor perception benchmark
3.14 3D vision (scans, reconstruction, point clouds)				
ScanNet [97]	3D	RGB-D	Large	3D reconstruction + semantic labels
ShapeNet [98]	3D	Meshes	Large	3D models for representation learning

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Dataset	Area	Modality	Scale	Typical tasks / notes
ModelNet40 [99]	3D	CAD	12k	3D shape classification baseline
S3DIS [100]	3D	Point cloud	Large	Indoor 3D semantic segmentation
3.15 Remote sensing and aerial imagery				
DOTA [101]	Remote	Aerial	Large	Oriented object detection in aerial images
xView [102]	Remote	Aerial	1M objs	Large-scale overhead object detection
DeepGlobe [103]	Remote	Aerial	Large	Road/building segmentation challenges
3.16 Document AI / OCR				
IAM Handwriting [104]	DocAI	Scans	Large	Handwriting recognition
PubTabNet [105]	DocAI	Scans	500k	Table structure recognition
3.17 Vision–language (image–text)				
Visual Genome [106]	V+L	Images+Ann	108k	Objects/attributes/relations; region descriptions
VQA [107]	V+L	Images+QA	Large	Visual question answering benchmark
Conceptual Captions (CC3M) [108]	V+L	Image-Text	3M	Caption pairs for multimodal pretraining
Conceptual Captions (CC12M) [109]	V+L	Image-Text	12M	Larger caption dataset
Flickr30k [110]	V+L	Image-Text	31k	Image captioning + retrieval
MSR-VTT [111]	V+L	Video-Text	10k	Video captioning / retrieval
LAION-5B [112]	V+L	Image-Text	5B	Web-scale image-text pairs; foundation-model pretraining
3.18 Generative modeling and super-resolution				
CelebA-HQ [113]	Gen	Images	30k	High-quality faces for GANs
DIV2K [114]	SR	Images	2k	Super-resolution benchmark
FFHQ [76]	Gen	Images	70k	High-quality face dataset for generation and editing

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Dataset	Area	Modality	Scale	Typical tasks / notes
3.19 Robustness and distribution shift				
ImageNet-A [115]	Robust	Images	7.5k	Adversarially filtered natural images
ImageNet-R [116]	Robust	Images	30k	Renditions (art, cartoons); domain shift
ImageNet-Sketch [117]	Robust	Images	50k	Sketch domain shift evaluation
Stylized-ImageNet [118]	Robust	Images	Large	Style transfer augmentation; texture/shape bias studies
3.20 Medical imaging (CV-specific)				
BraTS [49]	MedCV	MRI	2k+	Brain tumor segmentation benchmark
LUNA16 [119]	MedCV	CT	888	Lung nodule detection benchmark
DRIVE [120]	MedCV	Fundus	40	Retinal vessel segmentation (classic)

4 Conclusion and Outlook

This manuscript consolidates a representative set of datasets and benchmarks for machine learning and computer vision, with an emphasis on practical descriptors (modality, approximate scale, and typical tasks) and on benchmark hubs that standardize access and evaluation. The resulting overview can support rapid benchmarking, baseline selection, and clearer reporting by providing a shared map of commonly referenced datasets across subfields ranging from tabular learning and time series to vision–language and medical imaging. Importantly, the tables should be interpreted as entry points rather than prescriptions. The most appropriate benchmark depends on the intended deployment context, the available supervision, and the desired generalization regime.

Looking ahead, several trends are likely to shape benchmark practice. First, evaluation will increasingly combine multiple axes (accuracy, calibration, fairness, robustness, efficiency, and energy/cost) rather than focusing on a single metric. Second, multimodal and foundation-model evaluation will require more careful separation of pretraining sources and downstream benchmarks, including explicit tracking of contamination and deduplication. Third, dataset documentation and governance (licensing, consent, provenance, and privacy constraints) will become more prominent, especially for web-scale and healthcare data. Finally, living benchmark repositories and community-maintained leaderboards will continue to evolve, making dataset versioning and reproducible pipelines essential for credible comparisons. This overview is therefore intended to be updated over time and to encourage transparent, well-documented benchmark selection and reporting.

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